CYBERSECURITY INCIDENT AND COMPLIANCE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

AKSHAT SINGH

(241401008)

in partial fulfilment of the End Semester Examination for the course

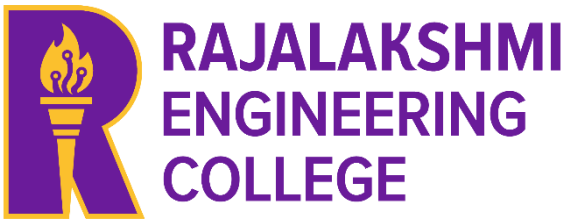
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEMS



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Cybersecurity Incident and Compliance Management System" is the bonafide work of Akshat Singh (241401008), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, **Akshat Singh (241401008)**, hereby declare that the project report titled "**Cybersecurity Incident and Compliance Management System**", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of work carried out by me individually. The Zoho Creator application described in this report was developed as an individual academic project, with AI-assisted application generation used where applicable and with implementation status documented honestly.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 03 November 2026

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AKSHAT SINGH

ABSTRACT

Cybersecurity teams need a reliable way to record incidents, assess severity, coordinate investigations, retain evidence, track vulnerabilities, and document remediation. When these activities are managed through separate spreadsheets, emails, or informal messages, incident information can become fragmented, duplicate entries may occur, ownership can be unclear, and management may not have a current view of unresolved risks. The Cybersecurity Incident and Compliance Management System (CICMS) is designed as a centralised low-code application to organise these activities in one workspace.

The project uses Zoho Creator to organise incident reporting and related modules for investigations, evidence, vulnerability tracking, remediation actions, and compliance assessments. Forms provide structured data capture, while lookup relationships are intended to connect related records. Reports and dashboard pages provide visibility into incident status and outstanding work. Zoho Creator workflows and Deluge scripts are intended to support incident identification, validation, severity-based notifications, duplicate prevention, and stakeholder alerts. During development, Zia-assisted workflow application encountered an internal validation error; therefore, those workflows are documented as requiring manual configuration and verification rather than being claimed as completed.

The application was reviewed using a staged test plan covering form access, data persistence, relationships, status transitions, notifications, duplicate prevention, dashboard accuracy, and permissions. The report distinguishes design objectives from features that must still be verified in the live application. No quantitative improvement is claimed because a controlled before-and-after measurement has not yet been recorded. The main outcome is a structured low-code prototype and a repeatable plan for completing and validating the incident-management lifecycle.

Keywords: Cybersecurity; Incident Management; Zoho Creator; Low-Code; Workflow Automation; Compliance.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Cybersecurity Incident and Compliance Management System (CICMS)
Application Domain	Cybersecurity incident response and compliance tracking
Student Name & Register No.	Akshat Singh – 241401008
Zoho Creator Edition	Zoho Creator edition: verify in account settings
Application Name (Link Name)	CICMS (confirm actual application link name)
Published / Live URL	https://akshatsingh2024csbsraj860.zohocreatorportal.in
Demo Credentials (per role)	Akshatsingh.20204.csbs@rajalakshmi.edu.in, Akshatsingh@123
Number of Forms / Subforms	7/2
Number of Reports (by type)	7
Number of Pages / Dashboards	1/1
Number of Workflows	4
Deluge Functions / Custom Actions	Configured Deluge workflows for incident assignment and status notifications
Integrations	No external integration confirmed in the current prototype
Roles & Profiles Defined	Customer,admin
Environments Used	Development environment (Zoho Creator)
Date of Production Deployment	2/11/2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PK	Primary Key
RBAC	Role-Based Access Control
SIEM	Security Information and Event Management
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho's AI assistant
CICMS	Cybersecurity Incident and Compliance Management System

CHAPTER 1
INTRODUCTION

1.1 Background

CICMS addresses the operational needs of a small or medium-sized organisation that must record cybersecurity events and coordinate response activities. The system is designed for security analysts, incident coordinators, compliance officers, and administrators. Its central process begins when a suspected event is reported and continues through triage, investigation, evidence collection, remediation, verification, and closure. A related compliance function records control assessments and links failed controls to corrective actions. The application is implemented as a low-code prototype on Zoho Creator; organisation-specific policies, production deployment, and integration choices depend on the final environment.

1.2 Problem Statement

Cybersecurity incident information is often distributed across email conversations, spreadsheets, and separate tracking tools. This makes it difficult to maintain a consistent incident history, assign responsibility, link evidence and remediation to the original incident, and provide timely status reports. CICMS proposes a centralised record-based application to improve traceability and coordination across the incident and compliance lifecycle.

- Incident details may be duplicated or incomplete across separate records.
- Manual assignment and follow-up can delay investigation and remediation.
- Managers may lack a consolidated, current view of incident status and compliance gaps.

1.3 Objectives

- To design structured forms and relationships for incident, investigation, evidence, vulnerability, remediation, and compliance records.
- To configure and test workflow automation for incident identification, validation, notifications, and lifecycle status changes.
- To provide reports and dashboard views that support monitoring of incident severity, status, and pending actions.
- To evaluate suitable Zoho Creator integrations and document any integration that is actually configured.
- To document access-control, testing, and deployment requirements and verify them before any production release.

1.4.Operational Context and Incident Categories

CICMS is intended to support a consistent response to events such as suspected phishing, malware alerts, unauthorised access, exposed credentials, unusual account activity, and policy violations. These examples describe the kinds of records the system may manage; they do not imply that the prototype detects these events automatically. A user or analyst must report the event, provide an initial description, and attach relevant contextual information.

Each incident record should answer five practical questions: what happened, when it was observed, which system or asset may be affected, how severe the potential impact appears,

and who is responsible for the next action. Consistent answers improve triage and reduce the need to reconstruct basic facts from separate conversations. The record should distinguish confirmed facts from assumptions and should be updated as the investigation progresses.

Incident categories and severity definitions should be documented before the application is used by multiple people. For example, severity may consider business impact, scope, sensitivity of affected information, service availability, and evidence of active exploitation. The final classification policy belongs to the organisation and must not be inferred solely from a dropdown selection.

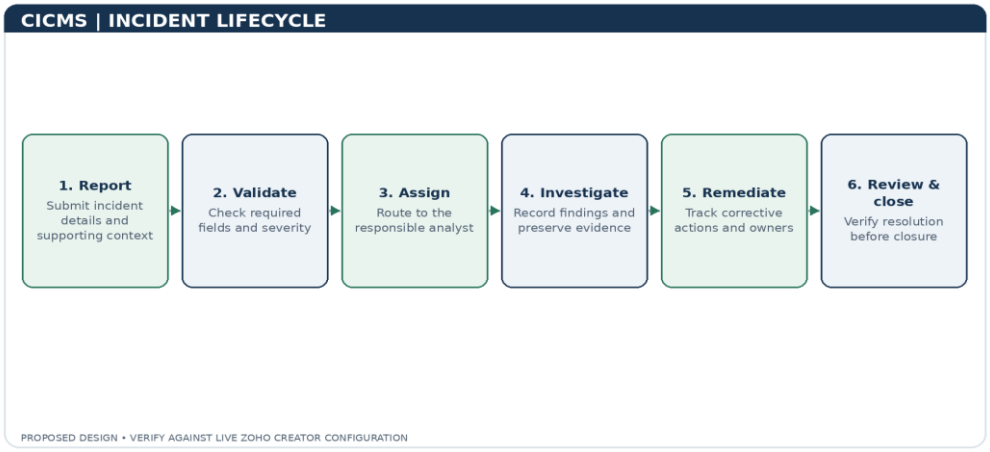


Figure 1.1: Operational incident lifecycle (proposed)

In scope: incident reporting; investigation tracking; evidence records; vulnerability tracking; remediation actions; compliance assessments; reports and dashboard design; workflow requirements; and access-control planning. Out of scope for the current prototype are autonomous incident containment, direct integration with a live Security Information and Event Management (SIEM) platform, guaranteed regulatory compliance, and production-grade deployment unless these are separately configured and tested.

1.5 Justification for a Low-Code Approach

Zoho Creator is suitable for a CICMS prototype because its visual form builder, lookup fields, reports, pages, and workflow tools reduce the amount of infrastructure and handwritten code required for a data-centric business application. The platform allows the developer to iterate on forms and user interfaces quickly and add Deluge logic where built-in configuration is insufficient. This approach is appropriate for an academic project because the data model and process can be demonstrated rapidly. Low-code does not remove the need for security design, testing, permission configuration, or review of generated automation [1], [2].

1.6 Organisation of the Report

Chapter 2 reviews the existing process and low-code context. Chapter 3 defines stakeholders and requirements. Chapter 4 presents the architecture, data model, workflow design, and access-control plan. Chapter 5 documents implementation status. Chapter 6 provides the test plan and results to be completed against the live application. Chapter 7 covers deployment and governance requirements. Chapter 8 discusses the prototype and its limitations, and Chapter 9 concludes with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a typical manual incident-management process, a user reports a suspicious event by email or message. A coordinator copies the details into a spreadsheet, assigns the case to an analyst, and follows progress through separate conversations. Evidence may be stored in shared folders, while remediation tasks are tracked independently. Closure is recorded after informal confirmation, and summary reports are assembled manually. This process can work at small scale but becomes difficult to audit as the number of incidents and stakeholders increases.



Figure 2.1: Existing Incident Handling Process

2.2 Limitations of the Existing System

- Incident records may be incomplete, duplicated, or distributed across separate files.
- Investigation, evidence, and remediation tracking may not share a consistent record identifier.
- Manual follow-up and report preparation reduce visibility into unresolved incidents and compliance gaps.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Requires custom frontend, backend, database, testing and deployment setup.	Visual forms/pages can be assembled quickly; workflows still require configuration and testing.
Skills required	Programming, database, hosting and security skills.	Visual app building plus Deluge and platform knowledge for automation.
Hosting & infrastructure	Developer or organisation configures hosting, database and deployment.	Hosting and platform infrastructure are provided by Zoho Creator.
Mobile application	Usually needs responsive web design or separate mobile development.	Creator offers mobile access, subject to configuration and plan.

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Security & access control	Custom authentication, authorisation and audit mechanisms must be implemented.	Platform permissions and roles are available but must be configured and tested.
Maintenance & change	Code changes, testing and deployment required.	Many form/report changes can be made visually; complex changes still require testing.
Customisation limits	Broad control over architecture and UI.	Faster development, but constrained by platform features and supported extension points.
Cost	Can require more engineering and infrastructure effort.	May reduce initial effort; subscription, feature and usage limits still apply.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation, n.d.	Zoho Creator Help Documentation	Describes form, report, workflow and application configuration.	Platform-specific implementation reference for CICMS.
2	Zoho Corporation, n.d.	Understand Lookup Field	Explains lookup-based relationships between forms.	Supports incident-to-investigation/evidence/remediation relationships.
3	Zoho Corporation, n.d.	Creating and Managing Form Workflows	Explains form workflow triggers and actions.	Supports event-based validation, record actions and notifications.
4	NIST, 2012	Computer Security Incident Handling Guide, SP 800-61 Rev. 2	Provides incident response phases and coordination practices.	Informs reporting, analysis, containment, recovery and lessons learned.
5	NIST, 2024	The NIST Cybersecurity Framework (CSF) 2.0	Organises cybersecurity outcomes, including Govern, Identify, Protect, Detect, Respond and Recover.	Provides a framework for linking incident response and compliance activities.
6	Zoho Corporation, n.d.	Configuring Data Access Actions for Form Workflows	Explains adding/updating related records from form workflows.	Supports planned linked-record creation and update actions.

2.5 Emerging Trends in Low-Code Development

Relevant low-code trends include AI-assisted application generation, visual data modelling, reusable workflow automation, and dashboard-driven monitoring. Zia can accelerate the initial creation of forms and pages, but generated components still require human review and functional testing. CICMS treats AI as an aid to development and, as a possible future enhancement, incident summarisation and triage support. Human analysts must remain responsible for severity decisions, containment, and closure.

2.7 Information Quality and Traceability

A centralised record model is useful only when the information entered into it is sufficiently complete, consistent, and traceable. CICMS should use stable identifiers for incidents and supporting records, standard values for status and severity, and lookup fields rather than free-text references wherever a relationship is required. A common identifier allows a reviewer to move from an incident to its investigation, evidence, vulnerability, or remediation action without relying on similar-looking titles.

Data quality controls should be proportionate to the prototype. Mandatory fields should be limited to information required for triage, while optional fields can capture details that become available later. Dates should use consistent date/time fields, status values should follow a controlled list, and evidence records should include a short description of provenance. Corrections to an existing record should preserve the meaning of the original event rather than silently replacing important investigative history.

These controls are design recommendations until their corresponding form properties, lookups, and workflows are configured and tested in Zoho Creator.

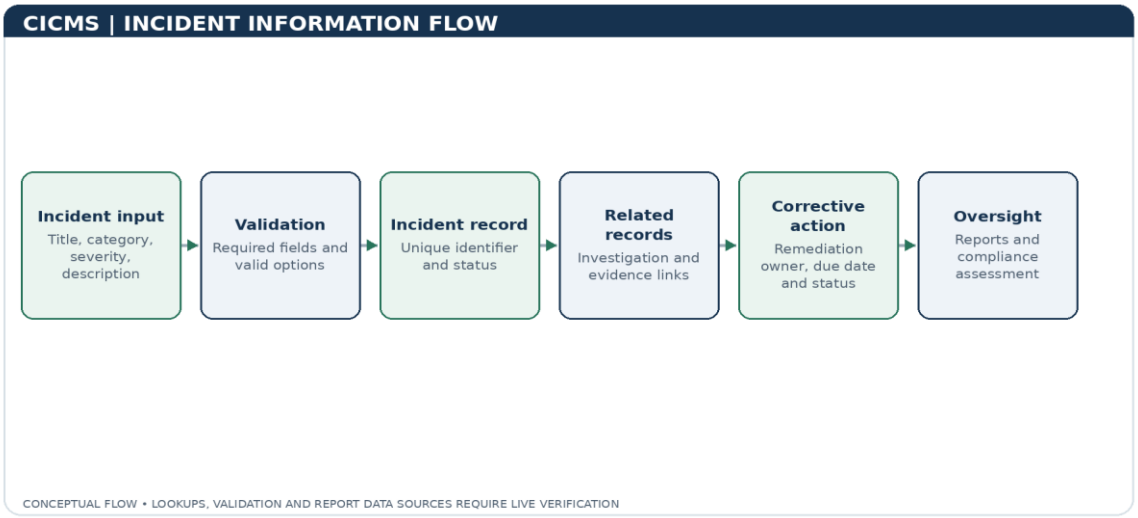


Figure 2.2: Information quality and traceability flow

CICMS provides a central workspace for registering incidents and maintaining linked investigation, evidence, vulnerability, remediation, and compliance records. Reports and dashboards are intended to expose open incidents, severity distribution, and pending corrective actions. Compared with separate spreadsheets, the proposed model provides a common record structure and clearer links between response activities. Workflow automation is part of the design, but the current implementation status must be verified because Zia's workflow validation failed and the five key workflows require manual configuration.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Application administrator	Maintain forms, roles, configuration and overall records.
Security Analyst	Incident response user	Register incidents, investigate events, attach evidence and update incident status.
Compliance Officer	Control-assessment user	Record compliance assessments, identify failed controls and track corrective actions.
Manager	Oversight and reporting user	Review dashboards, monitor high-severity incidents and review remediation progress.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall allow an authorised user to submit an incident with required details and save it as a record.	Security Analyst	High
FR-02	The system shall assign or store a unique incident identifier for each incident.	System / Administrator	High
FR-03	The system shall support linking an incident to investigation and evidence records.	Security Analyst	High
FR-04	The system shall track remediation actions and their status for incidents or vulnerabilities.	Security Analyst / Compliance Officer	High
FR-05	The system shall provide reports or dashboard views for incident severity, status and pending actions.	Manager / Administrator	Medium
FR-06	The system should notify configured stakeholders when defined severity or lifecycle conditions occur, once workflow setup is complete.	Security Analyst / Manager	Medium
FR-07	The system should prevent duplicate related records and block closure until required verification is complete, once rules are configured.	Administrator / Security Analyst	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports should load within an acceptable time for the expected academic prototype dataset; actual timings must be measured.
NFR-02	Security	Users should access only the records and actions permitted by their configured role.
NFR-03	Usability	Forms and reports should be understandable and usable through a supported browser; mobile use must be verified if claimed.
NFR-04	Availability	Application availability depends on the Zoho Creator service and the account/network environment.
NFR-05	Maintainability	Form names, field names, workflows and scripts should use consistent naming and be documented.
NFR-06	Auditability	Record changes should be traceable through available Creator history or audit features where enabled.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator account; edition to be verified in account settings
Scripting language	Deluge, only where scripts are configured
Browser	Current version of Google Chrome, Microsoft Edge, or Mozilla Firefox
API testing tool	Postman only if testing an API integration; no API integration is currently confirmed
Other Zoho services	No additional Zoho service confirmed as integrated
Mobile	Zoho Creator mobile app only if mobile access is tested
Client hardware	Computer or smartphone with internet access and a supported browser

3.6 Business Rules and Acceptance Criteria

Business rules translate the project objectives into observable behaviour. An incident must have a usable title, category, severity, and description before it can be accepted. A saved incident should be retrievable from the incident report. Supporting records should reference a valid parent incident where the relationship is required. A remediation action should identify an owner and a target date when those fields are part of the chosen form design.

Acceptance criteria should be phrased so that a tester can determine whether the requirement passes or fails. For example: given a valid incident, when the user submits the form, then the record is saved and appears in the report. Given a record missing a required field, when the user submits it, then the platform displays validation feedback and does not save an incomplete record. Given a Critical incident, when the post-save workflow runs, then the configured assignment and notification behaviour occurs exactly once.

Any acceptance criterion that depends on a workflow remains pending until the workflow has been manually configured and its execution has been observed. A planned rule is not evidence of an implemented feature.

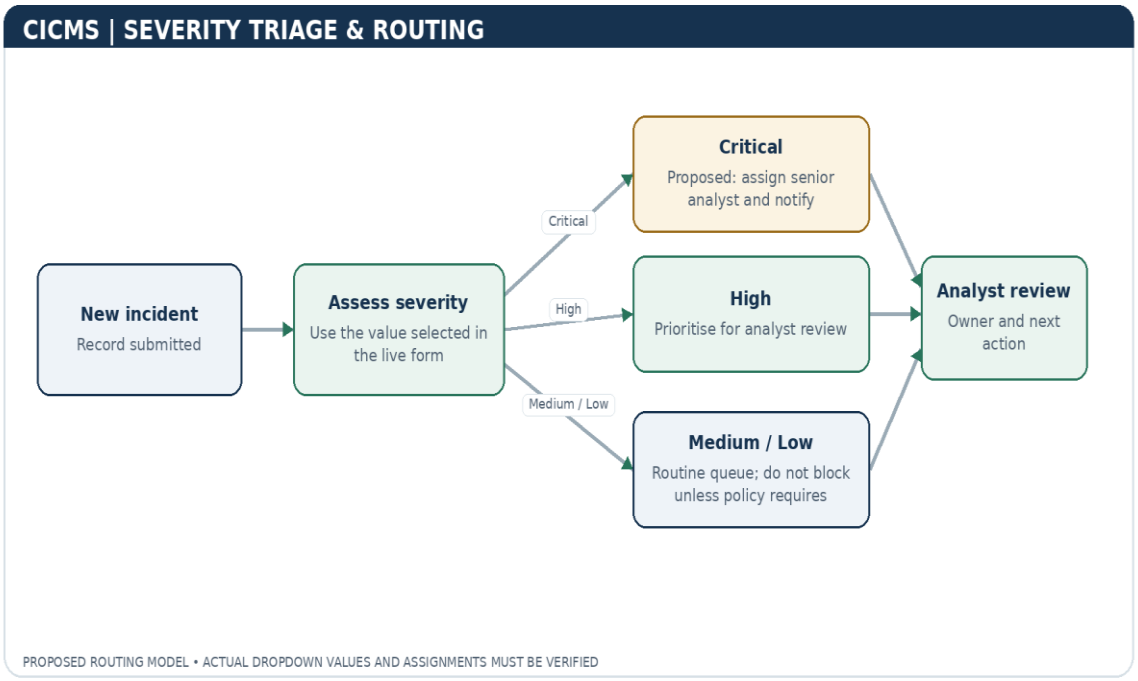


Figure 3.2: Severity triage and routing model (proposed)

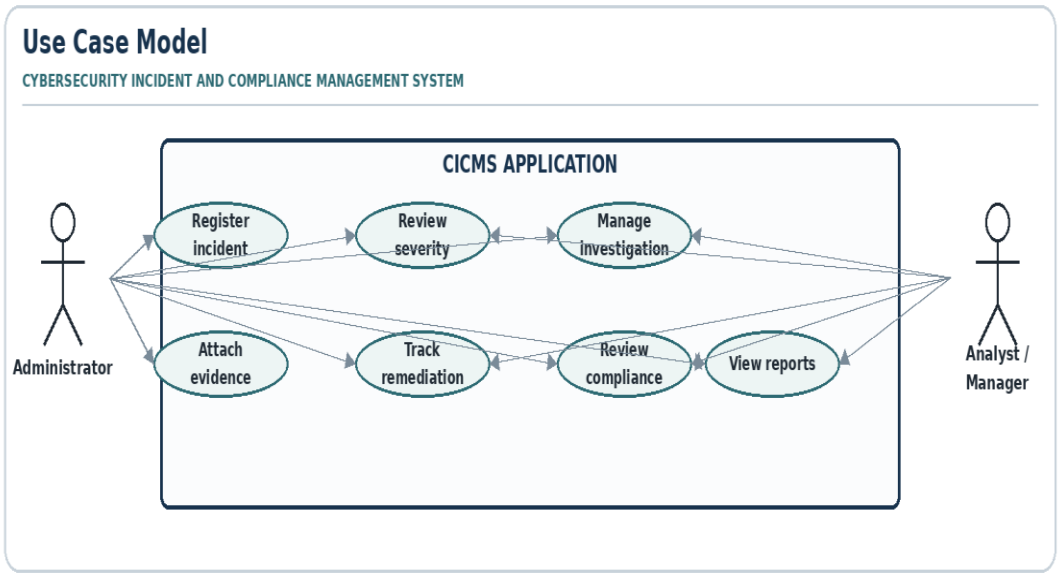


Figure 3.1: Use Case Diagram of CICMS

Table 3.5: Use Case Description – Report and Track an Incident

Use Case ID	UC-01
Use Case Name	Report and track a cybersecurity incident
Actor(s)	Security Analyst; Administrator
Pre-condition	User has access to the CICMS application and the incident form is available.
Main Flow	1. Open Incident Register. 2. Enter incident title, category, severity and description. 3. Submit the form. 4. Confirm the record appears in the incident report. 5. Link investigation/evidence records if those forms are available.
Alternate Flow	If required fields are missing, the form should display validation feedback. If the record does not save, the user records the error for debugging.
Post-condition	Incident record is saved and can be retrieved; related records are linked where supported.
Related FR	FR-01, FR-02, FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

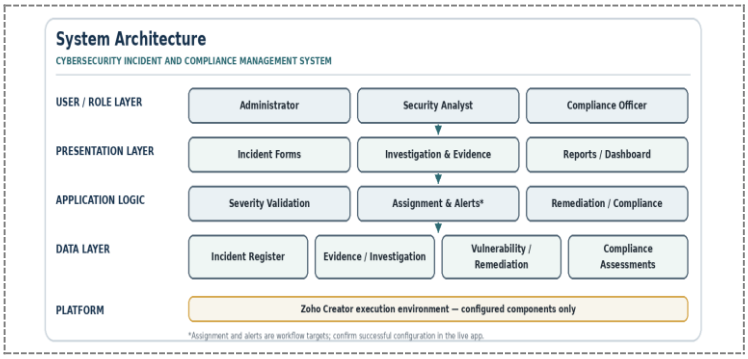


Figure 4.1: System Architecture of CICMS

Users access the application through Zoho Creator's web interface. Forms capture structured records; lookup fields connect related records; reports and pages present the saved information. Form workflows and Deluge scripts are intended to validate input, set identifiers, trigger notifications, and manage status transitions. The data layer consists of incident, investigation, evidence, vulnerability, remediation, and compliance records. External integrations are optional and are not claimed as active unless a connection has been configured and tested.

4.2 Data Model

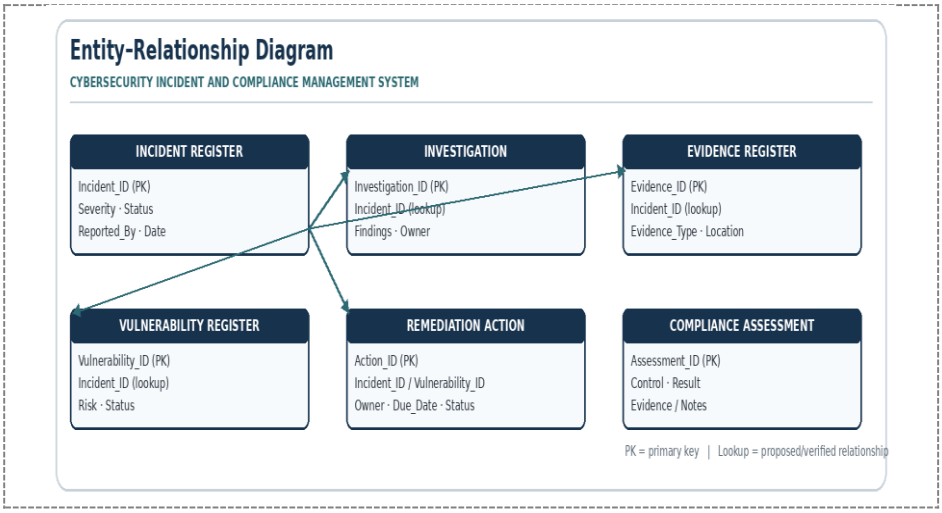


Figure 4.2: Entity–Relationship Diagram of CICMS

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Incident Register	Incident_ID, Title, Category, Severity, Status	Investigation, Evidence, Remediation	One-to-many	Lookup fields in child forms (verify in app)
Investigation Records	Investigation_ID, Incident, Investigator, Findings	Incident Register	Many-to-one	Incident lookup
Evidence Register	Evidence_ID, Incident, Evidence_Type, File, Timestamp	Incident Register	Many-to-one	Incident lookup; file upload if configured
Vulnerability Register	Vulnerability_ID, Asset, Risk, Status	Remediation Action Register	One-to-many (proposed)	Lookup in remediation form
Remediation Action Register	Action_ID, Related Incident/Vulnerability, Owner, Due_Date, Status	Incident Register / Vulnerability Register	Many-to-one	Lookup fields
Compliance Assessment	Assessment_ID, Control, Result, Assessor, Date	Remediation Action Register	Optional one-to-many	Lookup or linked corrective action if configured

4.3 Form Design

Table 4.2: Representative Field Specification – Incident Register

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Incident ID	Incident_ID	Auto Number / Single Line	Yes if generated automatically	Unique identifier; confirm actual field type and prefix	Use actual live field configuration
Incident Title	Incident_Title	Single Line	Yes	Required	Short incident summary
Category	Category	Drop Down	Yes	Phishing, Malware, Unauthorised Access, Other (if configured)	Confirm available choices
Severity	Severity	Drop Down	Yes	Low, Medium, High, Critical (if configured)	Used for prioritisation
Description	Description	Multi Line	Yes	Required	Do not include real credentials or sensitive personal data

4.4 Workflow Design Principles

Each workflow should have one clearly defined responsibility, a known trigger, explicit conditions, and a measurable result. Keeping validation, assignment, notification, duplicate prevention, and closure checks in separate workflows makes the configuration easier to test and troubleshoot. Where the platform permits, validation should occur before a record is saved, while email notifications should generally occur after a successful save so that the message refers to a persisted record.

Workflow conditions must use the actual field link names and field types from the live Creator form. For example, a dropdown field may be compared with a text value, but a user or lookup field may require a record or user reference rather than an email string. A script that assigns a text email address to an incompatible field can fail or produce an invalid assignment. Recipients should be configured using supported Creator user/email values and tested with a controlled account.

Workflow design should also consider repeat execution. Editing an incident can trigger another workflow run, so notification logic should avoid unnecessary duplicate emails. If a workflow updates a record, check whether that update can cause a second event. Keep test data separate from real records and document the final trigger, condition, action, and observed result for each workflow.

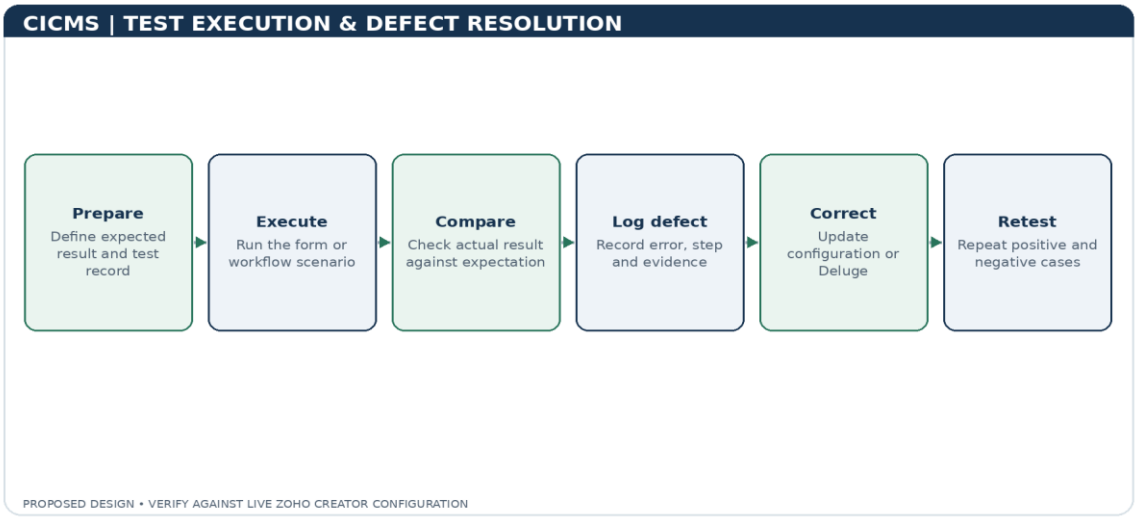


Figure 4.5: Workflow build and verification cycle

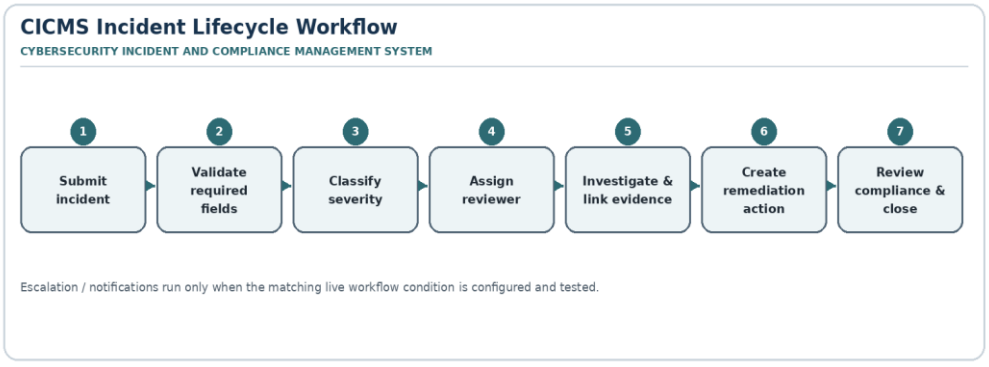


Figure 4.3: Proposed Incident Lifecycle Workflow

4.5 User Interface and Navigation Design

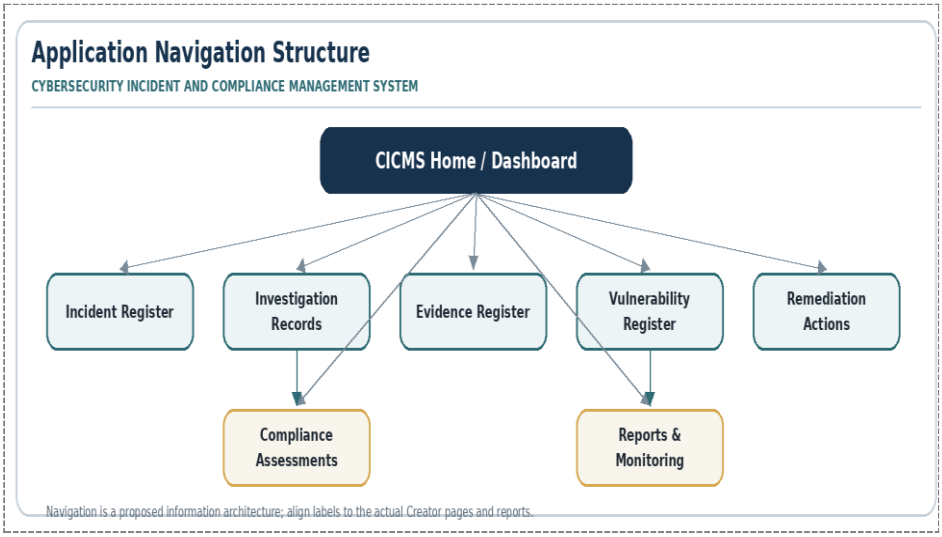


Figure 4.4: Proposed Navigation Structure and Dashboard Wireframe

4.6.1 Least-Privilege and Separation of Duties

Role-based access control should limit each user to the actions required for their responsibility. An administrator maintains application configuration and access settings; a security analyst records and investigates incidents; a compliance officer evaluates controls and tracks gaps; and a manager reviews summary information. These are proposed responsibilities and should be mapped to actual Creator permissions only after the roles and profiles have been configured.

Read access is not the same as edit access, and edit access is not the same as delete access. Evidence and investigation findings can contain sensitive information, so access should be limited to authorised users. The system should avoid placing secrets, passwords, authentication tokens, or unnecessary personal information in incident descriptions, screenshots, exports, and the report itself.

Permission testing should use separate test accounts or roles where possible. For each role, verify both allowed and disallowed actions: opening a permitted report, creating an allowed record, attempting an unauthorised edit, and confirming that restricted records or actions are not exposed. A role matrix in a design document is not proof that the platform enforces the same restrictions.

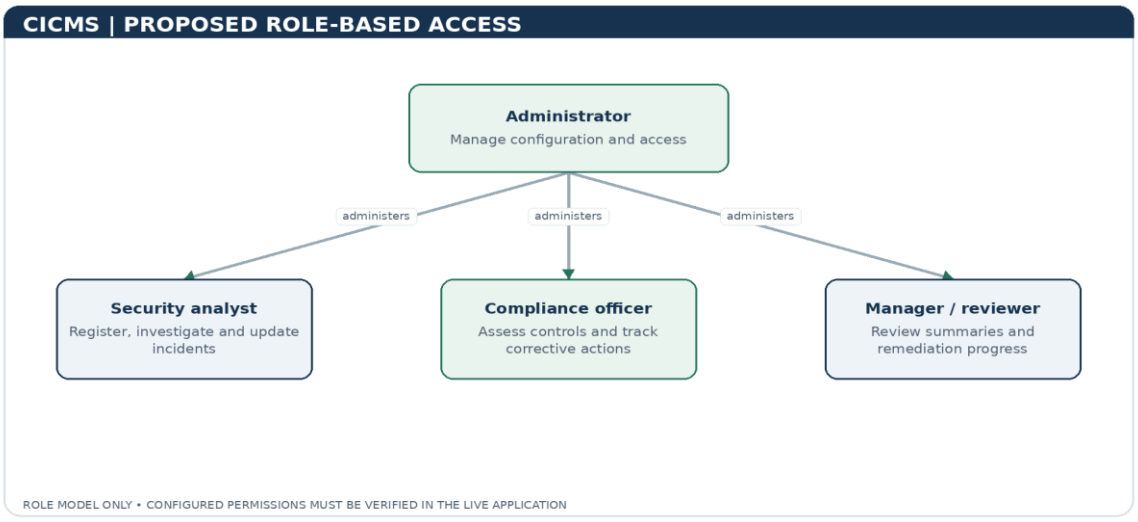


Figure 4.6: Proposed role-based access relationships

Table 4.3: Role–Permission Matrix

Component	Administrator	Manager	Security Analyst	Compliance Officer
Incident Register	A E V D	V	A E V	V
Investigation Records	A E V D	V	A E V	—
Evidence Register	A E V D	V	A E V	—
Remediation Action Register	A E V D	V	A E V	—
Compliance Assessment	A E V D	V	V A E	—

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The CICMS prototype was initiated using Zoho Creator's AI-assisted application-building capability. The generated application structure provides the starting point for forms, pages, and navigation. The workflow-generation stage encountered an internal validation error in Zoho Creator, so the workflow definitions must be created and tested manually. Before submission, the application name, actual form/report counts, live URL, and publishing status should be copied from the Creator environment rather than estimated.

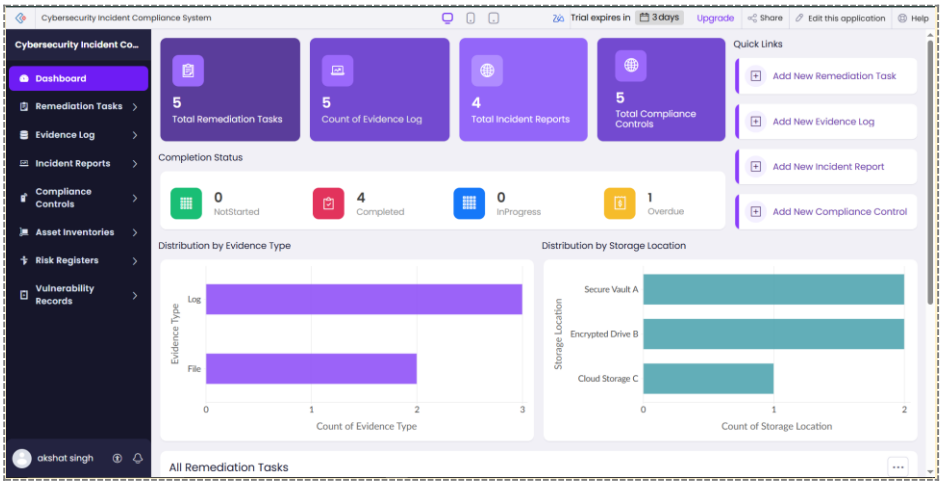


Figure 5.1: Application Home and Navigation Sections (Live Screenshot Required)

5.2 Forms and Data Capture

The intended data model contains an Incident Register as the main record and supporting forms for Investigation Records, Evidence Register, Vulnerability Register, Remediation Action Register, and Compliance Assessment. Common fields include a unique record identifier, title or summary, status, owner, timestamps, and links to related records. Lookup fields should be used wherever a supporting record must refer to its parent incident or vulnerability. The exact field labels, field types, link names, and mandatory settings must match the live application; Table 4.2 provides representative specifications to verify against the actual forms.

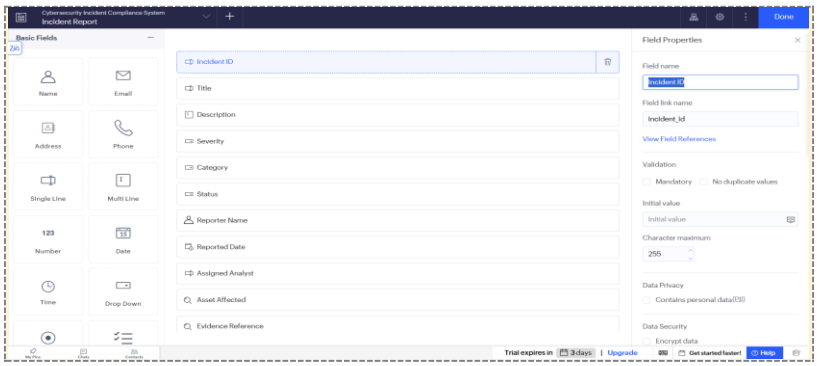


Figure 5.2: Incident Register Form in the Form Builder (Live Screenshot Required)

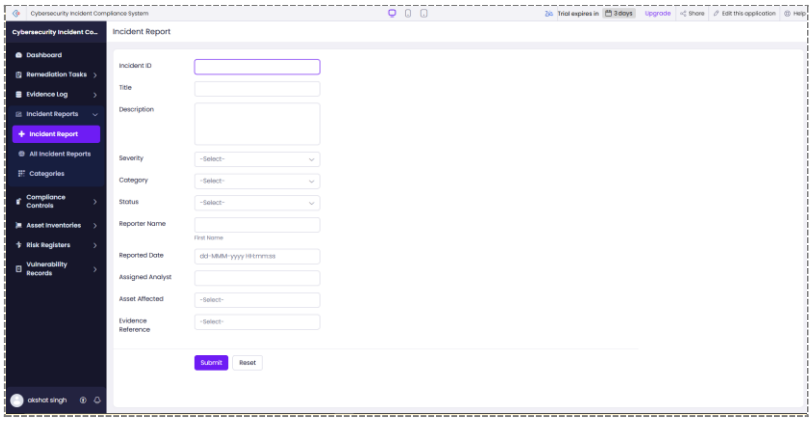


Figure 5.3: Incident Register Form in Live Mode (Live Screenshot Required)

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Incident Register Report	List	Incident Register	Filter by status/severity; sort by reported date descending	Open record / edit (if configured)	Administrator, Security Analyst
Investigation Tracker	List	Investigation Records	Group by related incident or investigation status	Open related incident (if configured)	Administrator, Security Analyst
Remediation Tracker	List / Kanban if configured	Remediation Action Register	Filter by owner/status; sort by due date	Update status (if configured)	Administrator, Security Analyst, Compliance Officer
Compliance Assessment Report	List / Chart if configured	Compliance Assessment	Group by control result and assessment date	Open corrective action (if configured)	Administrator, Compliance Officer
Incident Summary Dashboard	Chart / KPI if configured	Incident Register	Severity and status summaries; verify data source	Filter or open report (if configured)	Administrator, Manager

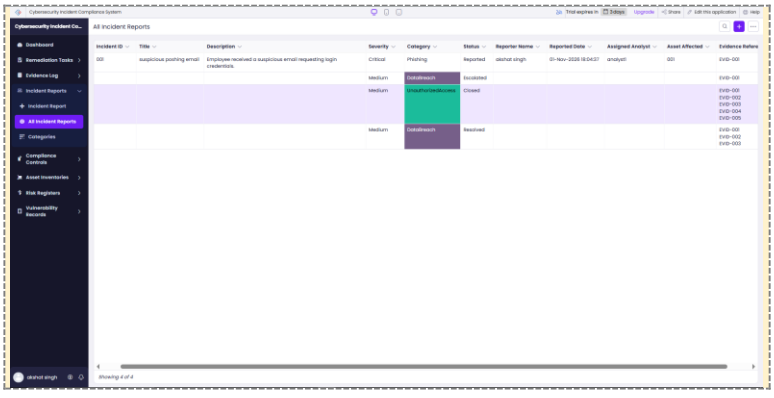


Figure 5.4: Incident Register Report (Live Screenshot Required)

5.4 Pages and Dashboards

The main page is intended to provide navigation to the incident and supporting modules, with a dashboard summarising incident counts, severity, status, and outstanding remediation. Each KPI or chart should be linked to a real Creator report or data source. The wireframe in Figure 4.4 is a proposed layout, not a screenshot of the live dashboard. Replace it with the actual page screenshot once the page is configured and populated with test data.

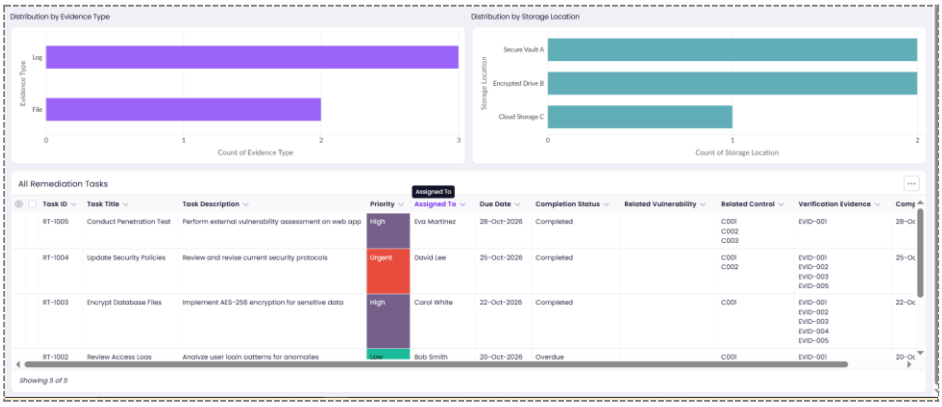


Figure 5.5: CICMS Dashboard (Live Screenshot Required)

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Incident Register → On Validate (planned)	Validate required fields and prevent invalid closure where possible.	if/else, validation, error feedback
DS-02	Incident Register → On Success (planned)	Generate or confirm a unique identifier and trigger severity alert after configuration.	field access, conditions, notification action
DS-03	Remediation workflow (planned)	Check for an existing related action before creating a new one.	criteria query, conditional logic
DS-04	Status-change workflow (planned)	Notify stakeholders when an incident is assigned, escalated or closed.	event condition, email/notification action
DS-05	Closure transition (planned)	Require investigation and verification to be complete before closure.	status checks, transition validation

- Code 5.1: Incident Status & Closure Validation

```
if(input.status == "Closed")
{
        alert "Incident cannot be closed without completing investigation and resolution.";
        cancel submit;
}
```

Code 5.2: Critical Incident Auto-Assignment

```
if(input.severity == "Critical")
{
        input.assigned_analyst = "akshatsingh.2024.csbs@rajalakshmi.edu.in";
}
```

No Deluge script is claimed as successfully deployed solely on the basis of the failed Zia workflow-generation attempt. The scripts listed in this report are proposed implementation tasks. When implemented, the validation script should check mandatory business rules; the notification workflow should send a message only for matching severity or status conditions; and duplicate prevention should search for an existing related record before creating another. Record the actual script link names, execution results, and any error messages after testing.

5.6 Recommended Manual Workflow Configuration Sequence

Review existing workflows before rebuilding anything. Verify each form, trigger, condition, field link name and Active/Suspended state in Zoho Creator. Preserve Workflow 4 as requested; inspect its purpose and behaviour before considering any change.

The discussed Critical rule assigns a senior analyst and emails a recipient when Severity is Critical; verify the user-field mapping and delivery. A separate script rejects Low and Medium submissions, so confirm this restriction matches policy. Stakeholder Alerts was proposed for edited incidents with Status = Escalated; verify that status option and the recipient address. Workflow 3's purpose is undocumented, and Workflow 4 should remain unchanged until inspected.

Record the outcome of every test: test record identifier, input values, expected result, actual result, date, and any error message. Keep the script version that passed testing and remove or disable obsolete duplicate workflows only after the replacement has been confirmed.

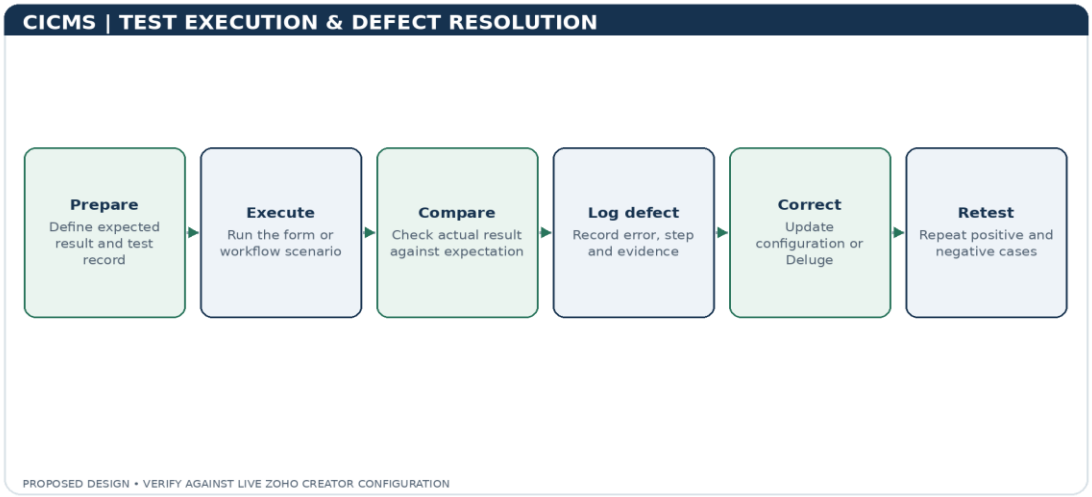


Figure 5.11: Recommended manual workflow rollout

Table 5.3: Workflow Configuration Status

Workflow	Trigger / condition	Intended action	Current evidence / next check
Critical assignment	Severity = Critical; verify trigger	Assign senior analyst; send email	Code discussed; test assignment and email.
Severity restriction	Severity = Low or Medium	Alert and reject submission	Code discussed; confirm this restriction is intended.
Workflow 3	Not documented in supplied notes	Do not infer action	Inspect live workflow and record its purpose.
Existing Workflow 4	Already created, per user	Preserve unchanged for now	Verify purpose and test result later.
Stakeholder Alerts	Edited incident; Status = Escalated	Email configured recipient	Confirm status option; replace placeholder email and test.

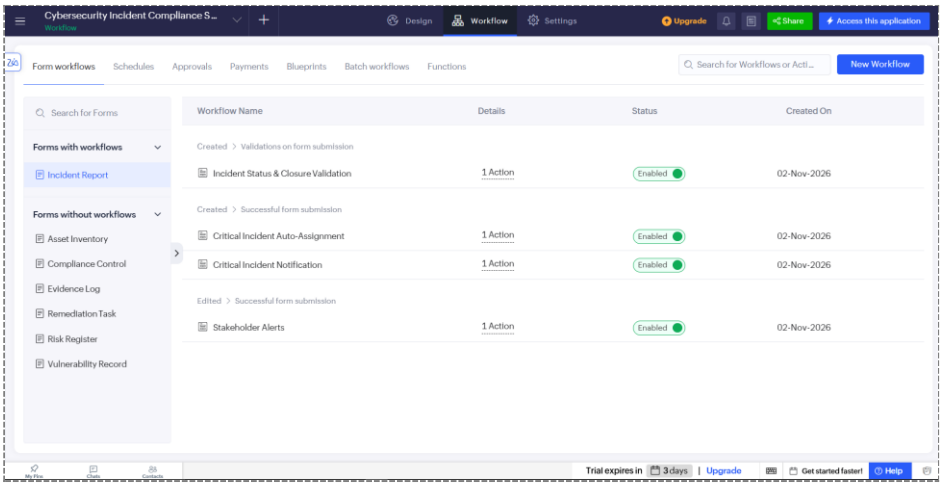


Figure 5.6: Workflow Configuration (Live Screenshot Required after manual setup)

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Zoho Creator native data	Creator forms and lookup fields	Native form relationships	Creator account permissions	Link incidents to investigations, evidence, vulnerabilities, remediation and assessments.
Email / notification action	Creator workflow notification	Built-in notification action	Configured recipient and permissions	Severity and lifecycle alerts; pending manual workflow setup.
External API	Not configured	Not applicable	Not applicable	No external API integration is claimed in the current prototype.

5.8 User Management and Role-Based Access

The intended roles are Administrator, Security Analyst, Compliance Officer, and Manager. Administrators maintain configuration and have broad access; analysts handle incident and investigation records; compliance officers maintain control assessments; managers review status and summary reports. This is a design-level role model. Only roles and permissions actually configured in Zoho Creator should be marked as implemented. Do not publish shared passwords in this report.

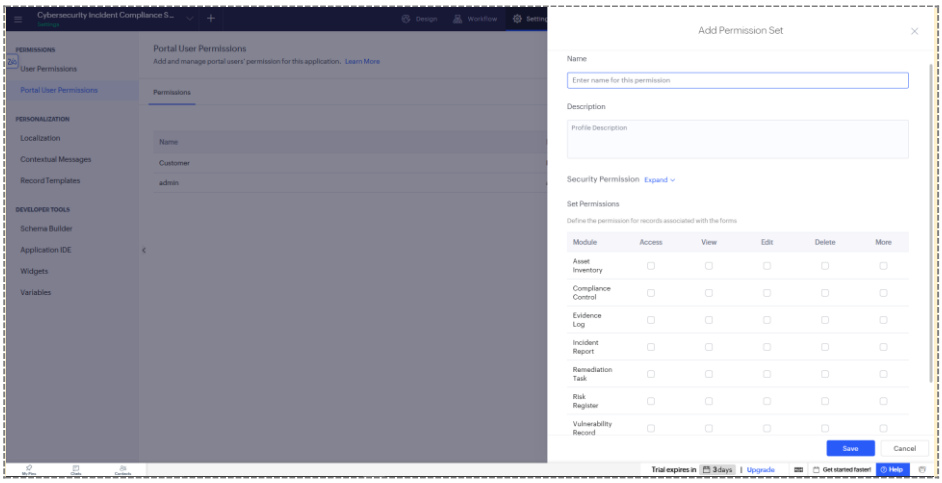


Figure 5.8: Roles and Permission Profiles (Live Screenshot Required)

5.10 Pro-Code Extension

A separate pro-code extension using Zoho Catalyst, custom JavaScript widgets, or external APIs has not been confirmed for the current prototype. Therefore, no such extension is claimed as implemented. If the final application remains within native Creator forms, reports, pages, and Deluge, this section documents that the project uses low-code configuration and does not require a separate pro-code service.

5.11 Mobile and Portal Access

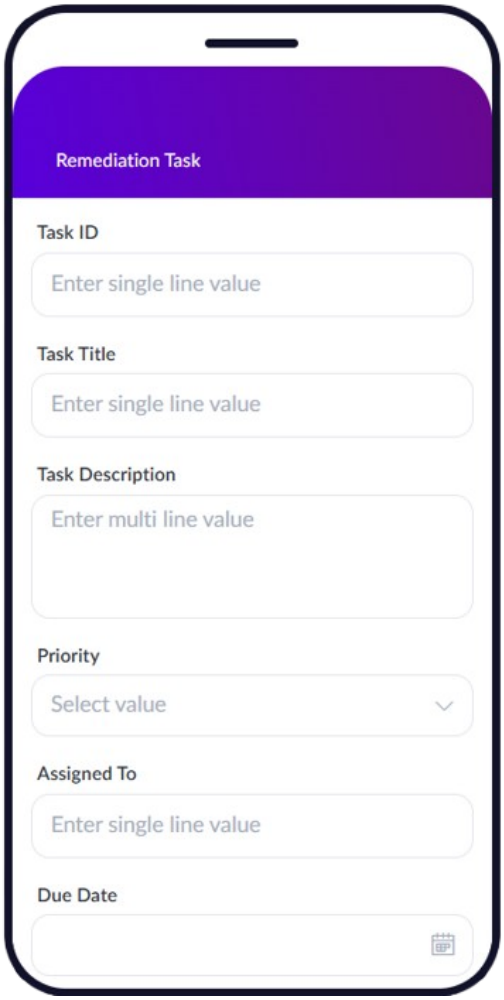


Figure 5.10: Mobile / Portal Access (Screenshot Only if Tested)

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is organised into form-level, integration, workflow, permission, and user-acceptance checks. Form-level tests verify mandatory fields and data persistence. Relationship tests verify that investigation, evidence, and remediation records point to the correct parent records. Workflow tests cover unique ID generation, closure validation, severity notifications, duplicate prevention, and stakeholder alerts. Dashboard tests compare displayed counts with the underlying reports. Because workflow generation previously failed, workflow cases must be marked Not Implemented or Fail until manually configured and successfully retested.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit incident with required fields	Enter fictional title, category, severity and description; submit.	Record created and visible in report.	To be tested	Not tested
TC-02	FR-01	Submit with required field missing	Leave a required field blank and submit.	Validation prevents incomplete record.	To be tested	Not tested
TC-03	FR-02	Create two incidents	Submit two separate incident records.	Identifiers are unique if ID workflow exists.	To be tested	Not tested
TC-04	FR-03	Link investigation to incident	Create investigation and select test incident in lookup.	Correct parent incident remains linked.	To be tested	Not tested
TC-05	FR-03	Link evidence to incident	Create evidence record and select test incident.	Evidence is linked to correct incident.	To be tested	Not tested
TC-06	FR-04	Create remediation action	Create action and link to incident/vulnerability.	Action saves with correct relation and status.	To be tested	Not tested
TC-07	FR-06	High-severity alert	Submit High or Critical incident after workflow configured.	Configured recipient receives alert.	Not testable until workflow configured	Not implemented
TC-08	FR-07	Closure validation	Attempt closure while investigation is incomplete.	Closure is blocked if rule is configured.	Not testable until workflow configured	Not implemented

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-07	Duplicate prevention	Edit same incident twice and inspect linked records.	No duplicate action is created.	Not testable until workflow configured	Not implemented
TC-10	FR-05	Dashboard totals	Compare dashboard counts with source report.	Counts match underlying records.	To be tested	Not tested
TC-11	NFR-02	Unauthorized role access	Log in with restricted test role and open restricted page.	Access follows configured permissions.	To be tested	Not tested
TC-12	FR-01	Edit existing incident	Change description and save.	Same record updates; no extra incident created.	To be tested	Not tested
TC-13	FR-05	Filter report by severity	Apply High/Critical filter.	Only matching incidents appear.	To be tested	Not tested
TC-14	FR-04	Update remediation status	Change status of a linked action.	Correct action updates and relation is preserved.	To be tested	Not tested
TC-15	NFR-03	Browser usability	Open key pages in supported browser and test navigation.	Pages render and navigation works.	To be tested	Not tested

6.3 Debugging Deluge Scripts

The first debugging step is to confirm that the relevant form workflow exists and is enabled. Next, verify the trigger event, condition, field mapping, and permissions. For Deluge scripts, use the editor's error output and info statements where supported. For workflow failures, inspect available execution logs and reproduce the issue with a single test record. The Zia workflow-generation error described during development occurred in the template-rendering/validation stage; it did not establish that the application forms or saved data were corrupted.

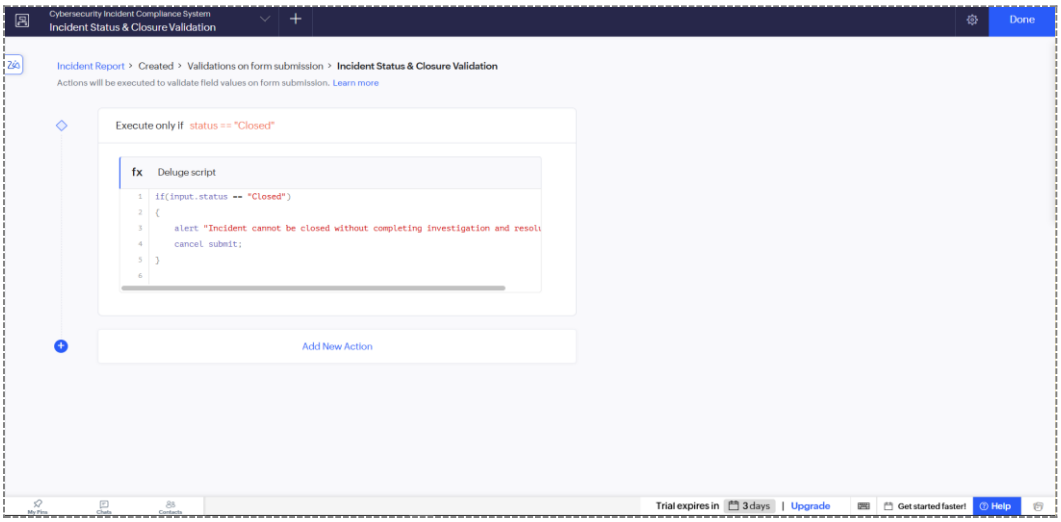


Figure 6.1: Workflow or Deluge Debug Output (Screenshot Required if available)

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Zia workflow generation stopped with template-rendering validation error.	Internal workflow validation exception reported by Creator; root cause not independently confirmed.	Manually configure each required workflow in the workflow builder and retest.	Open / workaround pending
BUG-02	Potentially missing workflow actions after failed generation.	Staged workflows were not applied; no form/field/data changes reported.	Inspect existing workflows and add missing ones individually.	Open until verified

6.6 Expanded Test Scenarios

Testing should cover normal operation, boundary conditions, invalid input, repeated events, and access restrictions. A successful save test checks persistence and report visibility; a validation test checks that missing mandatory values are rejected; a severity test checks that each allowed severity follows the intended path; and a notification test confirms both workflow execution and receipt by the intended mailbox. The test should also check that a non-Critical incident does not trigger a Critical-only alert.

Relationship testing should create a parent incident and then attempt to link investigation, evidence, and remediation records to it. The tester should verify that the linked record points to the correct incident and that filtering or navigation does not mix records with similar titles. Duplicate-prevention tests should repeat the same action and check whether the application creates a second record unexpectedly.

Security testing should include negative tests, not only successful access. A user who is not permitted to delete an evidence record should be unable to do so. A manager who is intended to have read-only access should not be able to change incident details. These outcomes must be verified in the configured application; they should not be inferred from the proposed permission matrix.

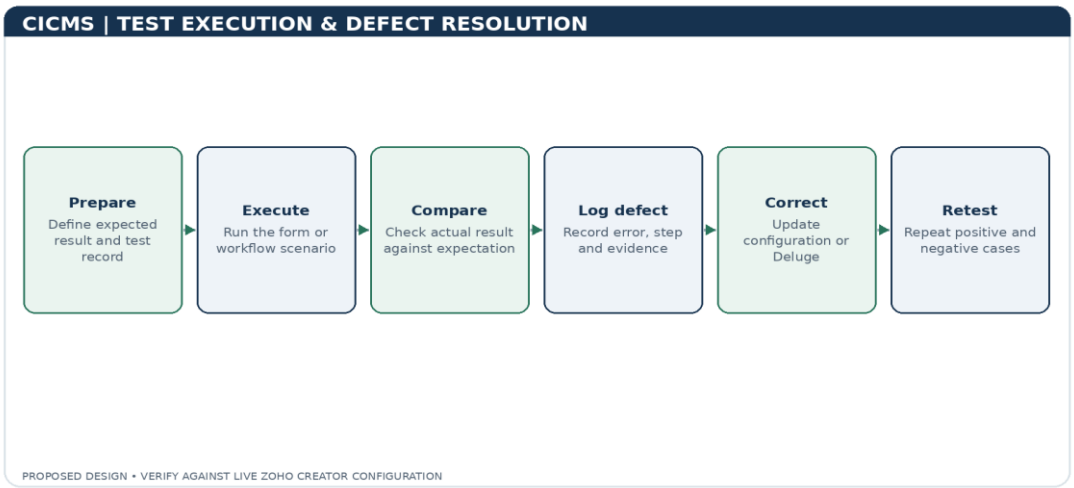


Figure 6.2: Test execution and defect resolution loop

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Tester 1 (to be filled)	Security Analyst	Complete after a real user tests incident entry and navigation.	Record actual feedback and changes.
Tester 2 (to be filled)	Manager / Compliance Officer	Complete after a real user tests reports and remediation tracking.	Record actual feedback and changes.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application should be developed and tested in a non-production state before publishing. Test records and workflows should be validated, permissions reviewed, and any required connections authorised before a production release. The current report does not assert that a production deployment has occurred; record the actual environment and publication date after checking the Creator application settings.

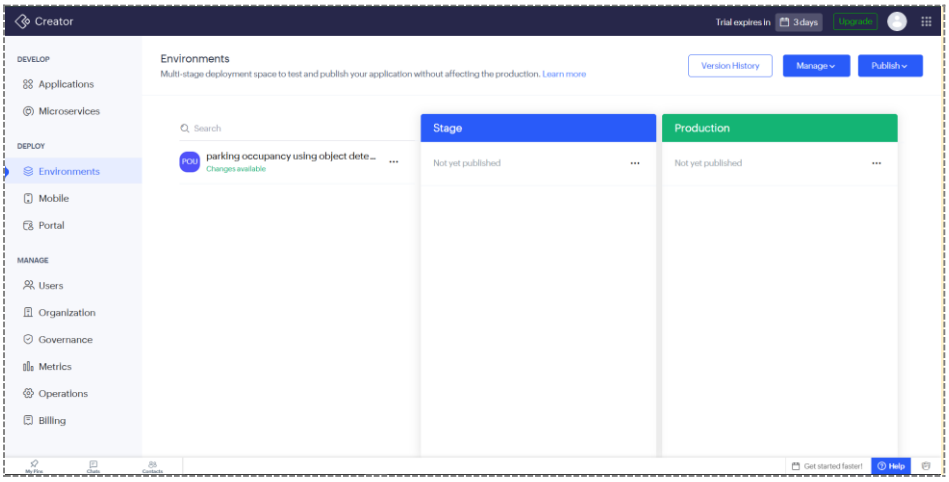


Figure 7.1: Deployment / Publishing Settings (Screenshot Only if published)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	Not yet verified
2	Sample/test data removed or isolated	Not yet verified
3	Roles, permissions and users configured in Production	Not yet verified
4	Connections authorised in Production	Not applicable until integration exists
5	Schedules and workflows enabled	Not yet verified; workflows pending manual configuration
6	Application published and URL verified	Not published / verify
7	Version / release notes recorded	To be recorded

7.2 Performance Optimisation

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Avoid repeated searches for related records	Planned workflow logic	Not measured	Not measured	Measure after implementation; do not claim improvement without data.
Use report filters for relevant records	Incident and remediation reports	Not measured	Not measured	Compare report load time on a known test dataset.
Use concise form layouts	Incident form	Not measured	Not measured	Check form completion time with a test user.

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permissions	Creator roles/profiles (verify)	Unauthorised access
Restricted evidence access	Evidence form/report permissions (verify)	Exposure of incident evidence
Secure credentials handling	Do not store secrets in plain text or screenshots	Credential leakage
Audit/history review	Creator audit/history feature if enabled	Untraceable changes

7.4 Governance and Administration

The Governance and Administration module of CICMS manages user roles, incident records, investigations, evidence, vulnerabilities, and remediation activities. It supports accountability by tracking incident statuses, assigning remediation tasks, and monitoring compliance assessments. Reports and dashboards help administrators review security activities and prioritize pending actions. Overall, the module provides a structured approach to cybersecurity management and improves traceability within the system.

7.5 Operational Governance and Record Retention

Governance defines who may change the application, how configuration changes are reviewed, and how long incident-related information is retained. A production implementation should identify an application owner, a backup administrator, a review cadence for permissions, and a documented process for restoring service after an accidental change. Workflow changes should be tested in a safe development or test environment before they are applied to operational records.

Evidence retention should follow the organisation's approved policy. Evidence records should capture what the item represents, who provided or collected it, when it was recorded, where the original is stored, and whether access is restricted. The prototype's file upload or attachment capabilities must be verified before the report claims evidence is securely retained. Sensitive evidence should not be copied into uncontrolled exports or shared screenshots.

Compliance assessment results should record the control being evaluated, the assessment date, the assessor, the outcome, supporting notes, and any corrective action. A failed control should be traceable to remediation, but a completed form alone does not establish legal or regulatory compliance. Compliance conclusions require the relevant control criteria, supporting evidence, and appropriate organisational review.

Governance practices for CICMS include consistent form and field naming, restricted administrator access, periodic review of roles, controlled changes to workflows, and export or backup of application data where supported. Audit or activity logs should be checked if enabled. The owner should document changes and retest critical workflows after edits. These controls should be recorded as implemented only after verification in the live application.

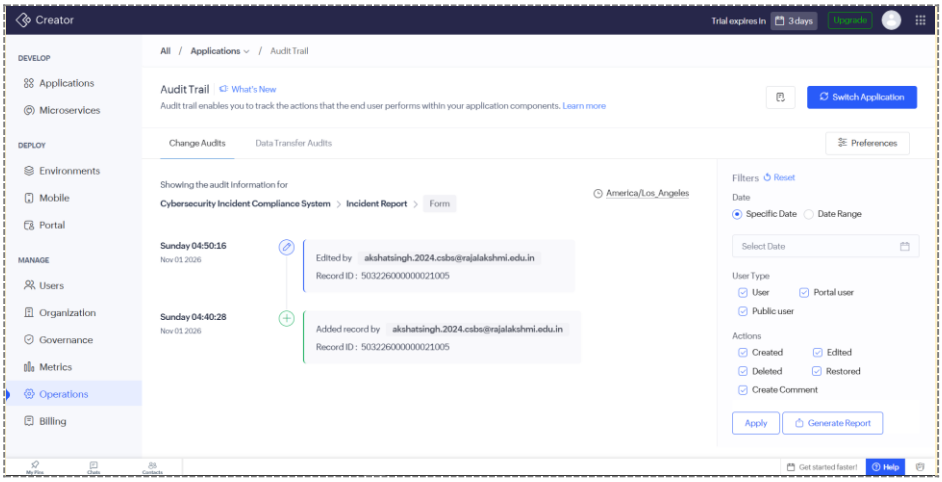


Figure 7.2: Audit Trail / Administration Settings (Screenshot Required if feature is enabled)

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The demonstration scenario begins with a fictional phishing report. The analyst records the incident and checks that it appears in the incident report. An investigation record and supporting evidence are linked to the incident. If a remediation action is required, the action is assigned an owner and tracked to completion. A reviewer verifies the result before closure. At present, the record-creation and relationship steps should be demonstrated only if they work in the live app; any automation that has not yet been manually configured must be identified as pending rather than shown as completed.

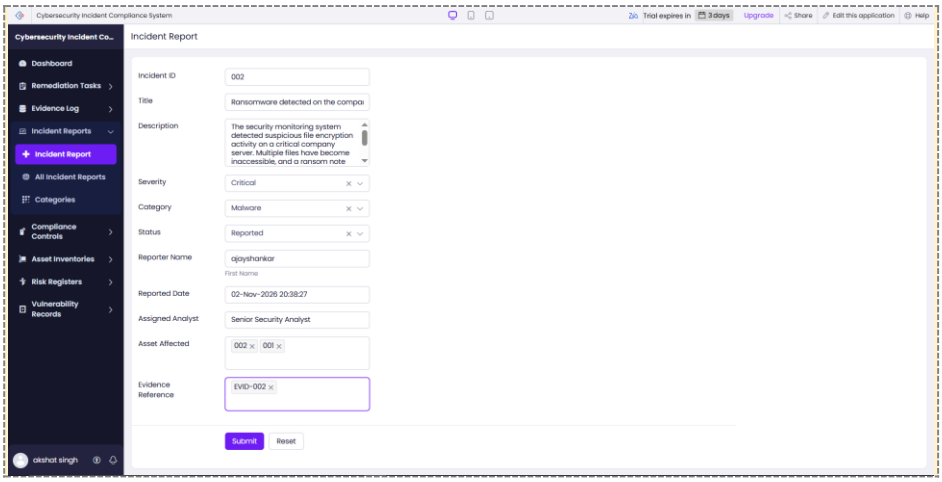


Figure 8.1: Scenario Step 1 – Incident Record Submitted (Live Screenshot Required)

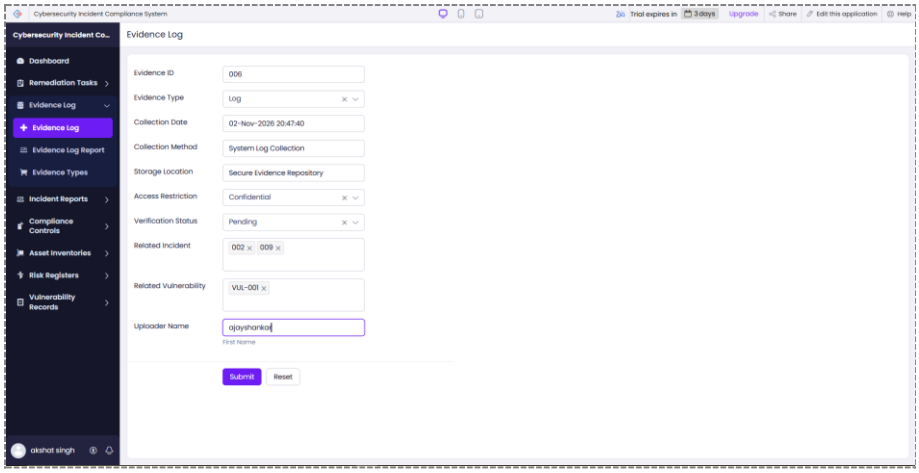


Figure 8.2: Scenario Step 2 – Related Investigation or Evidence Record (Live Screenshot Required)

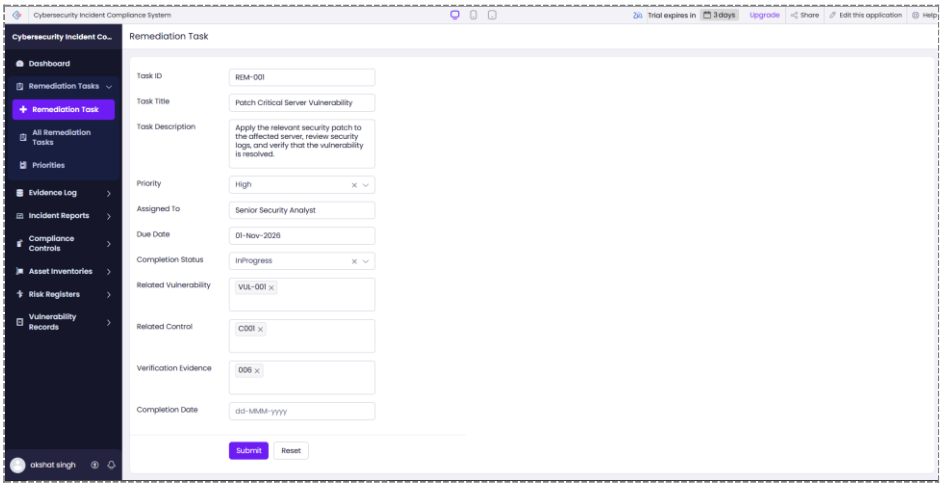


Figure 8.3: Scenario Step 3 – Remediation and Status Report (Live Screenshot Required)

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Structured incident and related-record model	Sections 4.2–4.3; forms and lookup fields	Verify live forms, fields and relationships.
Workflow automation	Sections 4.4 and 5.6	Manual workflow setup in progress. Code/configuration has been discussed for selected rules; verify saved state, field names, enabled status and test results in the live application.
Reports and dashboard	Sections 5.3–5.4	Verify counts against live report data.
Access control and governance	Sections 4.6 and 7.3–7.4	Verify actual role configuration before claiming complete.

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Incident processing	Details spread across email and sheets	Central incident record and linked supporting forms	Improved centralisation if configured and adopted
Manual steps	Repeated manual copying and follow-up	Forms and planned workflow actions	Automation benefit depends on successful workflow implementation
Report generation	Manual compilation	Creator reports and dashboard pages	Faster visibility when reports use correct data sources
Data accessibility	Access managed through separate files	Centralised access through Creator roles	Better control if roles and permissions are configured

8.4 Limitations of the Low-Code Approach

The current limitations include dependence on the Creator platform, limited flexibility compared with fully custom-coded applications, and the need to manually validate AI-generated components. During development, the workflow validation engine raised an internal template-rendering exception, preventing Zia's staged workflows from being applied. The workaround is to create the required workflows individually in Creator's workflow builder and test each one. Direct SIEM integration, automated containment, and AI-driven response remain outside the verified scope.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Developing CICMS using Zoho Creator demonstrated how low-code platforms simplify application development through visual forms, workflows, and reports. Compared to no-code tools, low-code allows customization using Deluge scripts, while pro-code development offers greater flexibility through traditional programming. This project highlighted the balance between rapid development, customization, and scalability offered by low-code solutions.

8.6 Risk and Limitation Analysis

The principal technical risk in the current project is the difference between the proposed design and the live configuration. A diagram can explain how a workflow should behave, but it cannot prove that a Deluge script executes successfully. The workflow-generation error encountered during development makes manual configuration and repeatable testing especially important. The report therefore distinguishes proposed behaviour from verified implementation.

A second risk is data-quality dependence. Dashboards and reports are only as reliable as the records and filters that feed them. Incorrect severity values, missing parent lookups, duplicate records, or inconsistent status labels can make summary information misleading. Controlled dropdowns, validation, stable identifiers, and periodic review help reduce this risk.

A third risk is platform dependency. The prototype relies on the availability, plan limits, and supported features of Zoho Creator. Future changes to licensing, permissions, APIs, or workflow limits may affect how the application can be extended. Before production use, the organisation should review data classification, access policies, backup arrangements, recovery needs, retention requirements, and any external integration security.

The project spans no-code configuration for forms and pages, low-code implementation for workflow conditions and Deluge logic, and potential pro-code extensions for advanced integrations. The AI-assisted builder helped establish the initial application structure, but the workflow validation failure showed why developers must understand the platform's native configuration tools and testing process. Critical cybersecurity decisions should not be delegated to unverified automation.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

CICMS is designed to centralise cybersecurity incident and compliance records in a Zoho Creator application. Its proposed modules cover incident reporting, investigations, evidence, vulnerability tracking, remediation, and compliance assessment. The project demonstrates how low-code tools can provide a structured data model and interface for a multi-step business process. The current status is a prototype requiring verification of live form relationships and manual configuration of workflows before the full incident lifecycle can be claimed as operational.

9.2 Future Enhancements

- Implement and test the five core workflows, then consider a human-reviewed Zia-based incident summarisation and triage assistant.
- Add a secure integration with an approved incident alerting or ticketing service, subject to available APIs and permissions.
- Explore a pro-code extension for SIEM data ingestion, advanced analytics, or external threat-intelligence enrichment.

9.3 Learning Outcomes and Career Relevance

The development of CICMS provided practical experience in building cybersecurity management applications using Zoho Creator. It improved understanding of low-code development, form design, record management, user roles, and workflow automation. The project also strengthened knowledge of incident tracking, vulnerability management, evidence handling, and remediation monitoring. These skills are relevant to careers in cybersecurity, security operations, IT administration, and application development.

9.4 Recommended Completion Roadmap

The next implementation stage is to verify the live application inventory: form names, field link names, field types, lookup relationships, reports, pages, roles, and current workflow status. This inventory should replace estimates in the report. The second stage is to configure and test required-field validation and Critical incident assignment. The third stage is to configure notification delivery, duplicate prevention, and closure checks, recording the outcome of each test.

The fourth stage is to verify the report and dashboard data sources, filters, and permissions using representative test records. The fifth stage is to review access restrictions, evidence handling, and administrative controls. Finally, update the implementation tables and screenshots only when the corresponding feature has actually been configured and tested. Any incomplete item should remain explicitly marked as pending rather than being presented as a completed capability.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	CICMS application and module navigation	Yes	CICMS application and security-management forms	Ch. 5	Application screenshot
II	Incident Report form	Yes	Incident Register / Incident Report; incident details, severity and status	Ch. 5	Incident form screenshot
II	Investigation record form	Yes	Investigation Records; investigation details and related incident, if configured	Ch. 5	Investigation record screenshot
II	Evidence Register form	Yes	Evidence Register; evidence details, collection method and storage location	Ch. 5	Evidence record screenshot
II	Vulnerability Register form	Yes	Vulnerability Record; CVE, CVSS, exploitability, asset, deadline and verification status	Ch. 5	Vulnerability form screenshot
II	Remediation Action form	Yes	Remediation Action Register; task, priority, assignee, due date, status and related records	Ch. 5	Remediation form screenshot
II	Admin and Customer roles	Yes	User roles / profiles configured in the application	Ch. 5; §7.4	Role settings screenshot

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	2 – Medium (prototype; verify features)	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2 – Medium (prototype; verify features)	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2 – Medium (prototype; verify features)	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	2 – Medium (prototype; verify features)	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2 – Medium (prototype; verify features)	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C1 Incident Status & Closure Validation

```
if(input.status == "Closed")
{
	alert "Incident cannot be closed without completing investigation and resolution.";
	cancel submit;
}
```

C2 Critical Incident Auto-Assignment

```
if(input.severity == "Critical")
{
	input.assigned_analyst = "akshatsingh.2024.csbs@rajalakshmi.edu.in";
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open the application URL provided by the project owner. 2. Sign in using your own authorised Zoho account. 3. Open CICMS from the available applications. 4. Use the navigation menu to select the required module. Mobile access should be described only if it has been tested.

D.2 Instructions for Security Analyst

- Open the Incident Register and submit or update a record using the fields available to your role.
- Open the related report to verify the saved record. Link investigation, evidence, or remediation records where the corresponding lookup fields are available.

D.3 Instructions for Manager / Compliance Officer

- Open the Incident Register and submit or update a record using the fields available to your role.
- Open the related report to verify the saved record. Link investigation, evidence, or remediation records where the corresponding lookup fields are available.